

F2Z

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Abstract

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1 Introduction

1.1 Acknowledgements

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2 Preliminaries

Given a prime q we let $\pi_q : \mathbb{Z} \rightarrow \mathbb{F}_q$ be the ring homomorphism mapping an integer to its reduction modulo q . We further let π_0 denote the identity map on $\mathbb{Z}$, so that $\pi_0(x) = x$ for all $x \in \mathbb{Z}$. We let $\mathbb{F}_0 = \mathbb{Z}$.

Given a finite field $\mathbb{F}$ of characteristic q , let $a(X) \in \mathbb{Z}[X]$ be an irreducible polynomial over $\mathbb{F}_q$ **Albert: yes?**, so that $\mathbb{F} = \mathbb{Z}[X]/(q, a(X))$ **Albert: consider isomorphism, or assume**

all finite fields are represented like this. Let $\psi_{\mathbb{F}} : \mathbb{Z}[X] \rightarrow \mathbb{F}$ be the canonical homomorphism from $\mathbb{Z}[X]$ to $\mathbb{F}$ obtained by reducing modulo the ideal $(q, a(X))$, i.e. by reducing coefficients modulo q and then reducing the resulting polynomial modulo $a(X)$.

3 F2Z

Let $\ell \geq 0$ be a dimension parameter, and $m \geq 0$ a blocklength parameter. Let $\mathbb{F}$ be a field (not necessarily a prime field), and let $\mathcal{C} \subseteq \mathbb{F}^m$ be a linear code of dimension k and blocklength m over $\mathbb{F}$. Let $k \geq 0$ be a public input length parameter, and let $Q \in \mathbb{F}[Y_1, \dots, Y_k, Z_1, \dots, Z_\ell]$ be a constraint polynomial with coefficients in $\mathbb{F}$ and on $k + \ell$ variables. Consider the following relation [Albert: what name?](#):

$$\mathcal{R}_{\mathcal{C}, \mathbb{F}}^Q := \left\{ \left(\begin{array}{l} \mathbf{x} = \mathbf{y} \in \mathbb{F}^k, \\ \mathbf{y} = \mathbf{f} \in \mathbb{F}^m, \\ \mathbf{w} = \mathbf{f} \in \mathbb{F}^\ell \end{array} \right) \middle| \begin{array}{l} \text{Enc}_{\mathcal{C}}(\mathbf{f}) = f \\ Q(\mathbf{y}, \mathbf{f}) = 0 \end{array} \right\}$$

When Q and $\mathbf{y}$ are such that $\mathbf{y} = (\mathbf{r}, \mu) \in \mathbb{F}^{\log \ell} \times \mathbb{F}$ and

$$Q(\mathbf{y}, \mathbf{f}) = \text{mle}[\mathbf{f}](\mathbf{r}) - \mu,$$

we denote $\mathcal{R}_{\mathcal{C}, \mathbb{F}}^Q$ by $\mathcal{R}_{\mathcal{C}, \mathbb{F}}^{\text{mle}}$.

Next let $\mathbb{F}_{\mathbf{c}}$ be a finite field of characteristic q , and let $\mathbb{F}_{\mathbf{e}}$ be either a finite field or $\mathbb{Q}$. For intuition, $\mathbb{F}_{\mathbf{c}}$ is the field over which we will commit our witness, and $\mathbb{F}_{\mathbf{e}}$ is the field over which we prove linear relations on the witness. We introduce the following short-hand notation:

$$\psi_{\mathbf{c}} := \psi_{\mathbb{F}_{\mathbf{c}}} : \mathbb{Z}[X] \rightarrow \mathbb{F}_{\mathbf{c}}, \quad \psi_{\mathbf{e}} := \psi_{\mathbb{F}_{\mathbf{e}}} : \mathbb{Z}[X] \rightarrow \mathbb{F}_{\mathbf{e}}.$$

Let $\mathcal{C} \subseteq \mathbb{F}_{\mathbf{c}}^n$ be a linear code of dimension k and blocklength n over $\mathbb{F}_{\mathbf{c}}$. Define [Albert: can we handle more general witness subsets?](#)

$$\mathcal{R}_{\mathcal{C}, \mathbb{F}_{\mathbf{c}}, \mathbb{F}_{\mathbf{e}}}^{\text{lin}} := \left\{ \left(\begin{array}{l} \mathbf{x} = (\mu, \mathbf{v}) \in \mathbb{F}_{\mathbf{e}} \times \mathbb{F}_{\mathbf{e}}^\ell, \\ \mathbf{y} = \mathbf{f} \in \mathbb{F}_{\mathbf{c}}^m, \\ \mathbf{w} = \mathbf{f} \in [0 \dots q-1]^\ell \subseteq \mathbb{Z}^\ell \end{array} \right) \middle| \begin{array}{l} \text{Enc}_{\mathcal{C}}(\psi_{\mathbf{c}}(\mathbf{f})) = f \\ \langle \psi_{\mathbf{e}}(f), \mathbf{v} \rangle = \mu \end{array} \right\}$$

We provide an IOPP for $\mathcal{R}_{\mathcal{C}, \mathbb{F}_{\mathbf{c}}, \mathbb{F}_{\mathbf{e}}}^{\text{lin}}$ next.

Protocol assumptions Assume for simplicity that $\mathbb{F}_{\mathbf{e}}$ is a prime field. The general case (when $\mathbb{F}_{\mathbf{e}}$ is a non-trivial field extension or the field of rationals) is handled in ??.

Fix an element $g \in \mathbb{F}_{\mathbf{c}}$ with sufficiently large multiplicative order, to be determined later. Let $S_q = \psi_{\mathbf{c}}([0 \dots q-1])$, and define the following bivariate polynomial with coefficients in $\mathbb{F}_{\mathbf{c}}$:

$$P(Y, Z) = \sum_{s \in [0 \dots q-1]} L_{S_q, \psi_{\mathbf{c}}(s)}(Z) \cdot Y^s \in \mathbb{F}_{\mathbf{c}}[Y, Z],$$

where $L_{S_q, \psi_c(s)}(Z)$ is the Lagrange polynomial associated to $\psi_c(s)$ over the set S_q , with degree at most $q - 1$. Note

$$P(Y, \psi_c(s)) = Y^s \quad \forall s \in [0..q-1].$$

We assume prover and verifier have access to an IOR $\prod_{\mathcal{R}_{\mathcal{C}, \mathbb{F}_c}^Q \rightarrow \mathcal{R}_{\mathcal{C}, \mathbb{F}_c}^{\text{eval}}}$ [Albert: define in prelims](#) that reduces $\mathcal{R}_{\mathcal{C}, \mathbb{F}_c}^Q$ to $\mathcal{R}_{\mathcal{C}, \mathbb{F}_c}^{\text{eval}}$, where

$$Q(Y_1, \dots, Y_{\ell+1}, Z_1, \dots, Z_\ell) = \prod_{i=1}^{\ell} P(Y_i, Z_i) - Y_{\ell+1}.$$

We further assume that prover and verifier have access to an IOPP for $\mathcal{R}_{\mathcal{C}, \mathbb{F}_c}^{\text{eval}}$.

Construction 3.1. The verifier receives $\mathbf{x} = (\mu, \mathbf{v}) \in \mathbb{F}_e \times \mathbb{F}_e^\ell$ and oracle access to $\mathbf{y} = f \in \mathbb{F}_c^n$. The prover receives $\mathbf{w} = \mathbf{f} \in [0..q-1]^k \subseteq \mathbb{Z}^k$ such that $(\mathbf{x}, \mathbf{y}, \mathbf{w}) \in \mathcal{R}_{\mathcal{C}, \mathbb{F}_c, \mathbb{F}_e}^{\text{lin}}$.

1. The prover sends $\mu' \in \mathbb{Z}$ such that $\langle \mathbf{f}, \psi_e^{-1}(\mathbf{v}) \rangle = \mu'$.
2. The verifier checks that $\psi_e(\mu') = \mu$.
3. Prover and verifier define $\gamma_i = \psi_e^{-1}(\mathbf{v}_i)$ for $i \in [\ell]$, and set to prove the following claim:

$$\prod_{i \in [\ell]} g^{\gamma_i \cdot \mathbf{f}_i} = g^{\mu'}.$$

Note that this claim is equivalent to

$$\prod_{i \in [\ell]} P(g^{\gamma_i}, \psi_c(\mathbf{f}_i)) = \prod_{i \in [\ell]} \sum_{s \in [0..q-1]} L_{S_q, \psi_c(s)}(\psi_c(\mathbf{f}_i)) \cdot g^{\gamma_i \cdot s} = g^{\mu'}.$$

To prove the above equality, prover and verifier engage in the IOR $\prod_{\mathcal{R}_{\mathcal{C}, \mathbb{F}_c}^Q \rightarrow \mathcal{R}_{\mathcal{C}, \mathbb{F}_c}^{\text{eval}}}$ with inputs

$$\left(\mathbf{x}' = ((g^{\gamma_i})_{i \in [\ell]}, g^{\mu'}), \mathbf{y}' = f, \mathbf{w}' = \psi_c(\mathbf{f}) \right).$$

4. At the end of this IOR, they are left with the following claim to prove:

$$((\mu', \mathbf{r}'), f', \mathbf{f}') \in \mathcal{R}_{\mathcal{C}, \mathbb{F}_c}^{\text{eval}}.$$

Accordingly, the prover and the verifier engage in a IOPP to certify the validity of this last claim.

We now describe a GKR-based IOR reducing $\mathcal{R}_{\mathcal{C}, \mathbb{F}_c}^Q$ to $\mathcal{R}_{\mathcal{C}, \mathbb{F}_c}^{\text{eval}}$

Construction 3.2.

4 Scratch

Construction 4.1. The verifier receives $\mathbf{x} = (\mu, \mathbf{v}) \in \mathbb{F}_e \times \mathbb{F}_e^\ell$ and oracle access to $\mathbf{y} = f \in \mathbb{F}_c^n$. The prover receives $\mathbf{w} = \mathbf{f} \in [0 \dots q - 1]^k \subseteq \mathbb{Z}^k$ such that $(\mathbf{x}, \mathbf{y}, \mathbf{w}) \in \mathcal{R}_{\mathcal{C}, \mathbb{F}_c, \mathbb{F}_e}^{\text{lin}}$.

1. The prover sends $\mu' \in \mathbb{Z}[X]$ such that $\langle \mathbf{f}, \psi_e^{-1}(\mathbf{v}) \rangle = \mu'$.
2. The verifier checks that $\psi_e(\mu') = \mu$. Then, the verifier samples $\alpha \in [0, 2^N - 1] \subseteq \mathbb{Z}$ and sends α to the prover. Let $\psi_\alpha : \mathbb{Z}[X] \rightarrow \mathbb{Z}$ be the evaluation map at α , so that $\psi_\alpha(a(X)) = a(\alpha)$.
3. The prover and verifier engage in a PIOP to prove the following equality:

$$\prod_{i \in [\ell]} g^{\psi_\alpha(\psi_e^{-1}(\mathbf{v}_i)) \cdot \mathbf{f}_i} = g^{\psi_\alpha(\mu')}.$$